

EXPERIMENT 16

High Level Programming Extensions – Procedures

Aim:

To implement procedures using program in MySQL.

PROCEDURES:

A procedure is a subprogram that performs a specific action.

Creating a procedure

We use the CREATE PROCEDURE statement to create a new stored procedure. We specify the name of stored procedure after the CREATE PROCEDURE statement. The DELIMITER command is used to change the standard delimiter of MySQL commands (i.e. ;). As the statements within the routines (functions, stored procedures or triggers) end with a semi-colon (;), to treat them as a compound statement we use DELIMITER

Calling stored procedures(Executing a procedure)

In order to call a stored procedure, you use the following SQL command:

CALLstored_procedure_name();

Questions

1. Create a stored procedure ShowPatients() to display all records from the Patient table.
2. Create a stored procedure GetPatientLevel() that accepts a patient's age and returns the patient category:
 - Age > 60 → **SENIOR**
 - Age 18–60 → **ADULT**
 - Age < 18 → **CHILD**

The manual uses CREATE PROCEDURE and CALL procedure_name() to create and execute procedures.

OUTPUT:

1)

```
1 • USE Hospital1;  
2  
3 • DROP PROCEDURE IF EXISTS ShowPatients;  
4  
5 DELIMITER //  
6  
7 • CREATE PROCEDURE ShowPatients()  
8 BEGIN  
9     SELECT * FROM Patient;  
10 END //  
11  
12 DELIMITER ;  
13  
14 • CALL ShowPatients();
```

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
	Patient_ID	Patient_Name	Age	Gender	Disease	Phone_No
▶	P001	Arun Kumar	25	M	Fever	9876543210
	P002	Priya	22	F	Diabetes	9876543211
	P003	Rahul	30	M	Asthma	9876543212
	P004	Sneha	28	F	Migraine	9876543213

2)

```
21 • CREATE PROCEDURE GetPatientLevel(IN patient_age INT)
22 BEGIN
23     IF patient_age > 60 THEN
24         SELECT 'SENIOR' AS Patient_Level;
25
26     ELSEIF patient_age >= 18 THEN
27         SELECT 'ADULT' AS Patient_Level;
28
29     ELSE
30         SELECT 'CHILD' AS Patient_Level;
31     END IF;
32 END //
33
34 DELIMITER ;
35
36 • CALL GetPatientLevel(25);
```

result Grid  Filter Rows: Export:  Wrap Cell Content: 	
Patient_Level	
ADULT	

RESULT:

Thus the high level procedures programs are executed successfully